AERO ENGINE SERVICES LTD T-6 AIRTOURER - COMPLIANCE CHECK LIST

F.A.R. PART 23 SUB-PART A

Airworthiness Standards

Part 23 Sub-Para No.	Requirement	Compliance		Reference
23.1 (a)	Applicability	Yes	The T-6 Airtourer is designed to meet Normal, Utility and Acrobatic Requirements.	
23.1 (b)	Compliance to be shown.	Yes	This document and pertinent reports indicate compliance.	
23.3 (a)	Normal Category	Yes	Clearance at 1900 lbs A.U.W.	
23.3 (b)	Utility Category	Yes	Clearance at 1900 lbs A.U.W.	
23.3 (c)	Acrobatic Category	Yes	Clearance at 1800 lbs A.U.W.	
23.3 (d)	More than one category.	Yes	Requirements of each category met.	

F.A.R. PART 23 SUB-PART B

Flight

Part 23 Sub-Para No.	Requirement	Compliance		Reference
21.(a)	Proof of Compliance Wt & C.G.	Yes	Complied with by testing.	FTRG P6
21.(b)	Flight test tolerances	Yes	Tests within tolerance.	
23.	Weights & C.G. range	Yes	Established.	
25.(a)(b)	Weight Limits	Yes	Maximum and minimum weights established.	Flight Manual.
29.	Empty Weight & C.G.	Yes	Aircraft weighed and condition defined.	Flight Manual
31.	Removable Ballast	N/A	No ballast is required.	
33.(a)	Propeller Speed & Pitch Limits	Yes	Flight test has shown safe operation.	
33.(b)(c)	Fixed Pitch & Ungoverned Variable Pitch Propellers.	N/A		
33.(d)	Constant Speed Propellers	Yes	Governor and pitch settings maintain speed within limits.	FTRG P36
45.	General	Yes	Standard atmosphere used.	

F.A.R. PART 23 SUB-PART B

Flight

Part 23 Sub-Para No.	Requirement	Compliance		Reference
49.(a)(b)	Stall speed, landing configuration	Yes	Stall speed less than 61 lets.	
49.(c)(d)	Stall speed determination	Yes	Stall speeds have been determined.	FTRG P96, 97
51.(a)	Take-off A/C greater than 6000lb	NA		
51.(b)	Take-off A/C less than 6000lb	Yes	Exceptional skill not reqd. Elevator control adequate.	FTRG P128
61.(a)	Climb perf A/C greater than 6000lb	NA		
61.(b)	Climb performance auw less than 6000lb	Yes	Climb is adequate in T.O. configuration	FTRG P113
67.	Climb: one engine inoperative	NA		
75.(a)	Landing auw greater than 6000lb	NA		
75.(b)	Landing auw less than 6000lb	Yes	Landings may be made safely	FTRG P124
77.(a)	a.u.w. greater than 6000lb	NA		
77.(b)	Balked landing auw less than 6000lb.	Yes	Climb in landing configuration adequate.	FTRG P109

Flight

Part 23 Sub-Para No.	Requirement	Compliance		Reference
161.(a)	Trim - General	Yes		
161.(b)	Lateral & directional trim	Yes	Aircraft is trimmed	
161.(c)	Longitudinal trim	Yes	Aircraft may be trimmed in all req'd conditions	FTRG P129 & 147
161.(d)	Multi-engine trim	NA		
171.	Stability general	Yes	Requirements are met	
171.(a)	Longitudinal stability	Yes	Control forces exist as req'd.	
171.(b)	Free return speed	Yes	Friction is very low. Friction band was not determined.	FTRG P130, 145
171.(c)	Variation in stick force with speed.	Yes	Stick force varies as req'd.	FTRG P130
175.(a)	Stability in climb	Yes	Stable longitudinally in climb.	FTRG P144
175.(b)	Stability in cruise	Yes	Stable longitudinally in cruise.	FTRG P144
175.(c)	Cruise with U/C down	NA	Fixed U/C.	
175.(d)	Stability in approach configuration	Yes	Stable	FTRG P144
177.(a)	Directional & lateral stability	Yes		FTRG P150-154

Flight

Part 23 Sub-Para No.	Requirement	Compliance		Reference
177.(b)	Directional & lateral stability	NA	Two control aircraft only.	
181.	Dynamic longitudinal oscillation	Yes	SPPO heavily damped.	
201.	Stall test procedure	Yes	Procedure used as stated in flight test.	
203.(a)	Controllability near stall	Yes	Controllability is maintained up to stall.	FTRG P137, 158
203.(b)	Turning flight stalls	Yes	No undesirable handling characteristics	FTRG P141, 163
203.(c)	Limited elevator control stalls	NA	T6 has adequate elevator control	
205.	One-engine-inoperative stalls	NA		
207.(a)(b)(c)	Stall warning device	Yes	Stall warning provided by L.E. sensor and horn	
221.(a)(b)(c)	Spinning	Yes	Aerobatic category spin requirement met	FTRG P143, 166
221.(d)	Characteristically incapable of spinning	NA		
231.(a)	Landplane ground handling	Yes	No nose over tendencies displayed.	FTRG P125
231.(b)	Water handling	NA		

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Flight

Part 23 Sub-Para No.	Requirement	Compliance		Reference
233.(a)	Crosswind handling	Yes	Control maintained in crosswinds.	FTRG P125
233.(b)	Power off landing controllability	Yes	No difficulties experienced with power off.	FTRG P125
233.(c)	Directional control in taxiing	Yes	Adequate directional control available.	FTRG P125
235.	Rough ground taxiing	Yes	Shock absorbing mechanism adequate	FTRG P125
251.	Vibration & buffeting	Yes	Aircraft is free of excessive vibration or buffet up to VD	FTRG P70-72
253.	High speed characteristics	NA	No VMO or MMO is established.	

F.A.R. PART 23 SUB-PART C

Structures

Part 23 Sub-Para No.	Requirement	Compliance		Reference
23.301 (a)	General	Yes	Strength requirements expressed as limit loads and ultimate loads.	
23.301 (b)	Balancing Inertia Forces	Yes	Inertia due to the mass of aircraft components has been considered in equilibrating air and ground loads.	
23.301 (c)	Deflections under Load	Yes	On no occasion was the deflection considered sufficient to change the load.	
23.301 (d)	Simplified Design Criteria	Yes	Appendix A used. Design A.U.W. is less than 6,000 lbs.	
23.303	Factor of Safety	Yes	A safety factor of 1.53 or higher has been used in structural tests. A safety factor of 1.50 has been used in theoretical analyses.	
23.305 (a)	Strength & Deformation - Limit	Yes	Structure supported limit test loads without detrimental, permanent deformation. Deformations at limit load do not interfere with safe operation.	

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Part 23 Sub-Para No.	Requirement	Compliance		Reference
23.305 (b)	Strength & Deformation - Ultimate	Yes	Structure supported ultimate loads for more than 3 seconds under static loading conditions.	
23.307 (a)	Compliance with Strength & Deformation requirements.	Yes	All major components have been subjected to static strength tests and/or theoretical analyses.	
23.307 (b)	Other testing.	Yes	As specified in appropriate para.	
23.321 through 23.459	Load Criteria	N/A	Covered by use of Appendix A as per A23.1(a).	
23.471	Ground Loads - General	Yes	In each specified ground load condition, the external reactions have been placed in equilibrium with the inertia forces in a rational or conservative manner.	
23.473 (a)	Ground Load Conditions & Assumptions	Yes	The Design Landing Weight is equal to the max. Take-off Weight.	REPORTS ANZ AT-007 ANZ AT-010 ANZ AT-012

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Structures

Part 23 Sub-Para No.	Requirement	Compliance		Reference
23.473 (b)(1)(2)(3) (c)(1)(2)	Ground Load Conditions & Assumptions - Design Landing Weight less than Max. Weight.	N/A		
23.473 (d)	Ground Load Conditions & Assumptions - Selected Limit Vertical Inertia Load Factor.	Yes	The selected limit vertical inertia load factor at the centre of gravity of the aeroplane is that which is obtained when landing at a vertical descent velocity of V = 8.77 ft/sec. ($V = 4.4(\frac{W}{S})^{\frac{1}{4}}$).	
23.473 (e)	Ground Load Conditions & Assumptions - Wing Lift During Landing.	Yes	The wing lift of $\frac{2}{3}$ the aircraft weight acting through the centre of gravity has been taken for all landing cases. The ground load reaction factor is equal to the inertia load factor minus the aerodynamic load factor of $\frac{2}{3}$.	REPORTS ANZ AT-007 ANZ AT-010 ANZ AT-012
23.473 (f)	Ground Load Conditions & Assumptions - Energy Absorption Tests.	Yes	See 23.725.	
23.473 (g)	Ground Load Conditions & Assumptions - Range of Inertia Load Factors.	Yes	The ground reaction factor is 2.78 and the c.g. inertia factor is 3.45.	Report ANZ-012 Page 5.

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Structures

Part 23 Sub-Para No.	Requirement	Compliance		Reference
23.477	Landing Gear Arrangement	Yes	The Airtourer has a conventional arrangement of main and nose gear.	
23.479 (a)(1)	Level Landing Conditions - Aircraft Attitude - Aeroplanes with Tail Wheels.	N/A	The Airtourer has a nose wheel.	
23.479 (a)(2)	Level Landing Conditions - Aircraft Attitude - Aeroplanes with Nose Wheels.	Yes	The appropriate attitudes for tricycle and two wheel landings have been considered.	REPORTS ANZ AT-007
23.479 (b)	Level Landing Conditions - Drag Components for Wheel Spin-up.	Yes	The drag load due to wheel spin-up was taken as 25% of the vertical load.	ANZ AT-010
23.479 (c)	Level Landing Conditions - Determination of Wheel Spin-up Loads.	Yes	The method set forth in Part 23, Appendix C has been used.	ANZ AT-012
23.481 (a)(1)	Tail Down Landing - Aircraft Attitude - Tail Wheel Aeroplanes	N/A	Airtourer does not have a tail wheel.	
23.481 (a)(2)	Tail Down Landing - Aircraft Attitude - Nose Wheel Aeroplanes	Yes	The critical attitude used for tail down landing is the maximum angle allowing ground clearance by each part of the aeroplane.	

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Part 23 Sub-Para No.	Requirement	Compliance		Reference
23.481 (b)	Tail Down Landing Conditions - Direction of Ground Reaction.	Yes	This requirement has been complied with using method of Appendix C.	
23.483	One Wheel Landing Condition - Aircraft Attitude & Ground Reactions.	Yes	The requirements of this paragraph have been complied with.	REPORTS ANZ AT-007 ANZ AT-010
23.485	Side Load Conditions.	Yes	The requirements of this paragraph have been complied with.	ANZ AT-012
23.493 (a)(b)(c)	Braked Roll Conditions.	Yes	The drag reaction applied at the ground contact point is 0.8 x the vertical reaction.	
23.497 (a)(b)(1)(2)(3)	Supplementary Conditions for Tail Wheels.	N/A		
23.499 (a)(1)(2) (b)(1)(2) (c)(1)(2)	Supplementary Conditions for Nose Wheels	Yes	The requirements of this paragraph have been complied with.	REPORTS ANZ AT-007 & ANZ AT-010
23.505 (a)(b)	Supplementary Conditions for Ski-Planes.	N/A	The Airtourer is not equipped with skis.	

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Part 23 Sub-Para No.	Requirement	Compliance		Reference
23.511 (a)(b)(c)	Ground Load - Unsymmetrical Loads on Multiple-Wheel Units.	N/A	The Airtourer does not have dual wheels.	
23.521 (a)(b)(c)	Water Loads	N/A	The Airtourer does not operate on water.	
23.561 (a)	Emergency Landing Conditions - Protection of Occupants under Crash Conditions.	Yes	The Airtourer has been designed to the prescribed requirements.	
23.561 (b)(1)(2)	Emergency Landing Conditions - Use of Seat Belts & Ultimate Inertia Forces.	Yes	The seats and seat belts have been designed to the conditions of the appropriate A.N.O. requirements which cover the F.A.R. requirements.	
23.561 (c)(1)(2)(3)	Aircraft with Retractable Landing Gear.	N/A	The Airtourer has fixed gear.	
23.561 (d)(1)(2)	Turnover	N/A	A turnover of this aircraft would be improbable.	
23.561 (e)	Restraint of Cabin Equipment etc.	Yes	Covered by original A.N.O. design requirements.	

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Structures

Part 23 Sub-Para No.	Requirement	Compliance		Reference
23.571 (a)(b)	Fatigue Evaluation - Pressurised Cabin.	N/A	The Airtourer is not pressurised.	
23.572	Fatigue Evaluation - Wing & Associated Structure.	Yes	The stress levels in the wing are lower than on the original AT-100 which has satisfactory fatigue service experience.	Report ANZ AT-017

F.A.R. PART 23 SUB-PART D

Design & Construction

Part 23 Sub-Para No.	Requirement	Compliance		Reference
23.601	Design & Construction - Tests on Items of Questionable Detail Design.	Yes	All bought-out parts not adequately substantiated by the suppliers and all parts of questionable detail design, have been cleared by test or, as appropriate, by suitable analysis.	
23.603 (a)(1)	Materials & Workmanship - Suitability & Durability of Materials.	Yes	The suitability and durability of the materials used in the structure has been established by experience and by tests.	
23.603 (a)(2)	Materials & Workmanship - Use of Approved Materials.	Yes	The materials used have approved Specifications that ensure their having the strength and other properties assumed in the design data.	
23.603 (b)	High Standard of Workmanship.	Yes		
23.605	Fabrication Methods.	Yes	Fabrication methods used ensure that consistently sound structures are produced. All processes are controlled by M.O.T.C.A.D. approved inspection.	
23.607	Self-locking Nuts - Prohibition of use on Bolts subject to Rotation.	Yes		

F.A.R. PART 23 SUB-PART D

Design & Construction

Part 23 Sub-Para No.	Requirement	Compliance		Reference
23.609 (a)(1)(2)	Protection of Structure Against Weathering & Corrosion.	Yes	A high standard of protective treatment is used throughout.	
23.609 (a)(3)	Protection of Structure Against Abrasion.	Yes	This requirement is complied with.	
23.609 (b)	Protection of Structure - Provision for Ventilation & Drainage.	Yes	Adequate provision is made in all components.	
23.611	Inspection Provisions.	Yes	Adequate provision for access to areas requiring recurring inspection, adjustments for proper alignment and functioning or lubrication have been made.	
23.613 (a)	Material Strength Properties & Design Values - Strength Properties.	Yes	The material strength properties used comply with this requirement.	
23.613 (b)	Material Strength Properties & Design Values - Design Values	Yes	Minimum specification properties have been used throughout.	

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Design & Construction

Part 23 Sub-Para No.	Requirement	Compliance		Reference
23.613 (c)	Material Strength Properties & Design Values - Use of Specified U.S. Materials Design Values.	Yes	The design values for all metallic materials and elements are those quoted in MIL-HDBK-5. No further non-metallic or composite design criteria was required.	
23.615 (a)(1)(2) (b)(c)	Design Properties - Conditions of Use.	Yes	Complies with the most conservative alternative provided, i.e. Minimum specification properties only are used at all times corresponding to MIL-HDBK-5 'A' values.	
23.619 (1)(2)(3)	Special Factors	N/A	As only well tried and proved manufacturing processes and inspection procedures are used, and the structure has been subject to a comprehensive series of strength tests, and has a high standard of protective treatment, no special factors other than those required have been used.	
23.621 (a)(b)(1)(2) (c)(1)(2)(3) (d)(1)(2)(3)	Casting Factors	N/A	There are no structural castings on the Airtourer.	
23.623 (a)	Bearing Factors - Clearance Fits Subject to Vibration.	N/A	No parts with clearance fits are subject to pounding or vibration in this aircraft.	

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Design & Construction

Part 23 Sub-Para No.	Requirement	Compliance		Reference
23.623 (b)	Bearing Factors - Control Surface Hinges & Control System Joints.	Yes	The control surface hinges andcontrol system joints comply with the requirements of 23.657 & 23.693.	
23.625 (a)(1)(2)(3)	Fitting Factors - Joints not Subject to Test.	N/A	All main components have been cleared by full scale tests to their critical conditions.	
23.625 (b)	Fitting Factors - Test Data for Joints.	Yes	Where test information was available the joint factor was neglected.	
23.625 (c)	Fitting Factors - Integral Fittings.	N/A	Integral construction is not used on the Airtourer.	
23.625 (d)	Fitting Factors - Seats, Safety Belts & Harness.	Yes	Original forward loading:- (AT-100) = 12g = 1.33 x 9g. The upward deceleration is covered by this 12g forward loading condition. Sideload design:- (AT-100) = 2.25 x g = 1.5 x 1.5. Therefore all fitting factors for seat belts and seats have been covered.	TR-02 Pg 145.
23.627	Fatigue Strength - Avoidance of Stress Concentration.	Yes	As far as practicable stress concentrations have been avoided.	

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Design & Construction

Part 23 Sub-Para No.	Requirement	Compliance		Reference
23.629	Flutter	Yes	Covered by flight testing to 23.629(c).	
23.641	Wings - Proof of Strength.	Yes	Wing strength is proved by full structural tests.	REPORTS ANZ AT-001, AT-008, AT-010 & AT-011.
23.651	Control Surfaces - Proof of Strength.	Yes	Limit and ultimate structural tests were carried out on all control surfaces.	REPORTS ANZ AT-003, AT-004, AT-005, AT-008, AT-010.
23.655 (a)	Control Surfaces - Installation Clearances.	Yes	No interference occurs between rudder and elevator.	
23.655 (b)	Control Surfaces - Installation - Adjustable Stabilizer.	N/A	The Airtourer does not have an adjustable stabilizer.	
23.657	Control Surfaces - Hinges	Yes	The hinges have been stressed for bearing strength.	REPORT ANZ AT-014
23.659 (a)(b)(c)	Control Surfaces - Mass Balance	Yes	The mass balances have been stressed or structurally tested to these requirements.	REPORT ANZ AT-006.
23.671 (a)	Control Systems - Control Surface Functioning.	Yes	Smoothness and ease of control movement demonstrated during operation test to 23.683.	

F.A.R. PART 23 SUB-PART D

Design & Construction

Part 23 Sub-Para No.	Requirement	Compliance		Reference
23.671 (b)	Control Systems - Arrangement & Identification.	Yes	All controls are arranged and identified in a logical manner.	
23.673 (a)	Primary Flight Controls - Definition.	Yes		
23.673 (b)	Primary Flight Controls - Two-Control Aircraft.	N/A	The Airtourer does not have a two-control system.	
23.675 (a)(b)(c)	Control Systems - Stops	Yes	Suitable stops are incorporated in all systems and these were tested to ultimate pilot effort loads during control system tests.	
23.677	Control Systems - Trim Systems	No.		
23.679	Control System Locks	Yes	A gust lock is provided which warns the pilot and cannot be engaged in flight.	
23.681 (a)	Control System Limit Load Static Tests	Yes	This requirement covered by structural static ultimate tests.	REPORTS ANZ AT-006 & AT-008.
23.681 (b)	Special Factor Requirements for Control System Joints Subject to Angular Motion.	Yes	See 23.693	

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Design & Construction

Part 23 Sub-Para No.	Requirement	Compliance		Reference
23.683 (a)(b)	Control System - Operation Tests.	Yes	Operation tests were carried out on all control systems to limit pilot forces defined in 23.397(b).	REPORTS ANZ AT-006 & AT-008.
23.685 (a)	Control System Details - Prevention of Jamming, Chafing etc.	Yes	The control systems have been designed and installed to prevent jamming, chafing and interference from cargo, passengers and loose objects.	
23.685 (b)	Control System Details - Prevention of Entry of Foreign Objects.	Yes	The detail design is such that the entry of foreign objects into places that would jam the control system is unlikely.	
23.685 (c)	Control System Details - Prevention of Cable or Tube Slapping.	Yes	A rubbing block prevents slapping of the elevator tube against other items.	
23.685 (d)	Control System Details - Minimising the Possibility of Incorrect Assembly.	Yes	The elements of the control system are either reversible or constructed to prevent incorrect assembly.	
23.687	Spring Devices	Yes	A spring device is used in the trim system but failure of a spring would result in only a small increase in the elevator control loads.	

F.A.R. PART 23 SUB-PART D

Design & Construction

Part 23 Sub-Para No.	Requirement	Compliance		Reference
23.689 (a)	Cable Systems - Approved Specifications.	Yes	Each cable, cable fitting, turnbuckle, splice and pulley is of an approved specification.	
23.689 (a)(1)	Cable Systems - Minimum Cable Diameter.	Yes	No cable used in the primary control circuit is less than $\frac{1}{8}$" diameter.	
23.689 (a)(2)	Cable Systems - Change in Cable Tension due to Operating Conditions & Temperature Variations.	Yes	Each cable system has been designed to produce an acceptable cable tension throughout the range of travel under operating conditions and temperature variations.	
23.689 (a)(3)	Cable Systems - Visual Inspection.	Yes	All fairleads, pulleys, terminals and turnbuckles can be visually inspected.	
23.689 (b)	Cable Systems - Matching of Pulleys & Cables, Cable Guards, Alignment of Cables in Pulleys.	Yes	These requirements have been complied with.	
23.689 (c)	Cable Systems - Change in Cable Direction.	N/A	No fairleads are employed in the control system.	
23.689 (d)	Cable Systems - Use of Clevis Pins	Yes	All clevis pins are retained with washers and split pins.	